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To the Editors-in-Chief,
Information Processing & Management

Dear Editors,

I submit the research article “How much retrieval gain reaches the answer? Transfer ratios and retrieval-stage cost of open embedding models in retrieval-augmented question answering” for consideration in Information Processing & Management.

Dense retrievers for retrieval-augmented generation are usually chosen by their retrieval scores. The manuscript asks how much of a paired retrieval gain reaches the answer. On Natural Questions and HotpotQA, it compares BM25 and six open embedding models under two 7–8B readers with corpus, prompt and decoding held fixed, and reports a paired transfer ratio, the exact-match gain divided by the Hit@5 gain, with bootstrap intervals. On NQ, dense encoders passed 0.16–0.27 exact-match points per Hit@5 point, and a registered replication on the full 2.68-million-passage corpus agreed for the encoders it covered. On HotpotQA, the ratios were lower once retrieval was scored by complete evidence. The paper also reports the measured retrieval-stage cost of each encoder on a free GPU.

The work fits the journal’s interest in the evaluation of information retrieval and question-answering systems. Its predictions were registered in dated design documents before the second and third study phases, all deviations are listed, and every reported number is generated by script from the committed outputs. Code, prompts, raw outputs and analysis scripts are available in the repository linked on the title page.

The manuscript is original, has not been published, and is not under consideration elsewhere. It has not been posted as a preprint. I am the sole author and have no competing interests; the work received no specific funding. The use of generative AI tools is declared in the manuscript. In line with the journal’s double-anonymised review, the manuscript file contains no author information; author details are in the separate title page.

Thank you for considering this submission.

Sincerely,
Muhammad Adeel